Mohamad **Dikshie Fauzie**

Forward Deployed Engineer · Applied AI, Agentic Systems & Cloud Solutions

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Summary

Senior Systems Engineer and Applied AI Solutions Architect with 10+ years of mission-critical production experience bridging enterprise customer requirements with production-ready software architectures and AI-driven solutions. Holds a Ph.D. in Distributed Systems from Keio University (supervised by Internet Hall of Fame founder Prof. Jun Murai) and a B.Sc. in Physics from ITB. Proven track record in forward-deployed technical delivery: engaging directly with enterprise clients and financial participants, translating ambiguous business challenges into scalable application workflows, and rapidly driving prototypes into resilient production systems. Combines hands-on Python/API engineering, Generative AI application development, enterprise RAG pipelines (chunking, hybrid retrieval, reranking), and agent orchestration with full-lifecycle MLOps/LLMOps observability (Prometheus, Grafana). Certified in NVIDIA AI Infrastructure, AWS Cloud, and Google Project Management. Tokyo resident with business Japanese (JLPT N2 level) and fluent English, fully prepared for high-impact on-site customer engagement.

Skills

Applied AI & GenAI	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration (LangChain, LlamaIndex), Prompt Engineering, Semantic Search, Vector Databases (FAISS, Qdrant), Model Evaluation & Benchmarking
Application Eng.	Python (FastAPI, Flask, AsyncIO, PyTest), JavaScript/Node.js, RESTful APIs, Microservices Architecture, Event-Driven Systems, Modular Design, Git CI/CD, Containerization (Docker)
Cloud & LLMOps	AWS (Certified), Kubernetes, Linux (RHEL, Ubuntu), MLOps/LLMOps Lifecycle (Deployment, Versioning, Monitoring, Rollback), AI Observability (Prometheus, Grafana, OpenTelemetry), Logging & Metrics
Systems & Data	Distributed Systems, Dell PowerScale (Isilon) Parallel Storage, PostgreSQL, Redis, NetBox (DCIM), High-Throughput Telemetry Ingestion, Data Security, Access Control (RBAC), Audit Governance
Client Delivery	Forward Deployed Engineering, Customer-Facing Solutions Architecture, Requirements Translation, Rapid Prototyping to Production, Stakeholder Alignment, Google Certified Project Management
Languages	English (Fluent / Business Required), Japanese (Business Proficiency / JLPT N2 level, 15+ yrs in Tokyo), Indonesian (Native)

Work Experience

Japannext Co., Ltd. (Largest Proprietary Trading System / PTS in Japan)

Tokyo, Japan

Forward Deployed Systems Architect & Technical Delivery Lead (Co-Manager) — *Oct. 2014 - Present*

- Served as primary customer-facing technical lead for institutional participants, broker-dealers, and technology partners, translating complex connectivity and data processing requirements into production-ready technical architectures.
- Architected and deployed Python-based microservices, REST APIs, and automated telemetry ingestion pipelines, eliminating manual operational reporting and achieving 100% automated SLA compliance reporting.
- Built operational telemetry and incident trend analytics platforms, analyzing high-volume event logs and ticket patterns to optimize engineering resource allocation, accelerate MTTR, and achieve 50% power distribution balancing across high-density systems.
- Designed and deployed open-source DCIM inventory tracking (NetBox) and real-time observability pipelines (Prometheus, Grafana), accelerating fault isolation and providing deep operational visibility across multi-datacenter environments (TY3, CC2).
- Spearheaded end-to-end datacenter migrations under tight power and operational constraints; executed seamless cutovers with zero downtime 2 weeks ahead of schedule, unlocking 50% system expansion capacity.
- Architected high-throughput parallel data pipelines on Dell PowerScale (Isilon) storage and automated LTO backup workflows, gaining 50% I/O throughput for multi-terabyte daily data logs while maintaining strict FSA regulatory data governance and security controls.

Keio University - JICA JST Indo Project

Tokyo, Japan

Research Assistant (Distributed Systems, Algorithms & AI) — *Apr. 2011 - Sep. 2014*

- Researched distributed network topologies, peer-assisted CDNs, and probabilistic gossip protocols, modeling complex network dynamics and information dissemination under node churn.
- Engineered high-performance simulation models in Python and C++, evaluating graph dynamics and energy optimization; published 3 peer-reviewed papers in IEEE and IEICE conferences and journals.

Keio University

Tokyo, Japan

System Administrator (Campus Core Infrastructure) — *Oct. 2007 - Sep. 2014*

- Administered campus-wide core infrastructure (BGP/OSPF routing, DNS, email, directory services, and firewall security), ensuring 99.99% uptime for research laboratories and academic faculty.

Applied AI & Engineering Projects

Enterprise Retrieval-Augmented Generation (RAG) Intelligence Pipeline

Python, LangChain/LlamaIndex, Qdrant, Docker

Lead Architect & Developer — *2024*

- Designed and deployed an end-to-end RAG system enabling semantic search and conversational question-answering over complex technical manuals, API documentation, and regulatory operational logs.
- Engineered modular document ingestion pipelines featuring semantic chunking, dense vector embeddings, hybrid retrieval (BM25 + dense vector search), and cross-encoder reranking to optimize context recall and precision.
- Implemented automated evaluation frameworks measuring answer faithfulness, context relevance, and hallucination rates, establishing repeatable benchmarks for model quality assurance.

Autonomous Multi-Agent Task Orchestration & Diagnostic Workflow

Python, FastAPI, OpenAI/Anthropic APIs, Pydantic

Architect & Developer — *2023 - 2024*

- Engineered a modular agent-based architecture with dynamic tool integration to automate complex diagnostic query workflows and multi-step data retrieval tasks.
- Implemented robust task orchestration logic, structured JSON schema validation, error recovery loops, and human-in-the-loop escalation gates to guarantee execution reliability in production workflows.

LLMOps Observability & Telemetry Pipeline

Python, Prometheus, Grafana, OpenTelemetry, Docker

Architect & Developer — *2023 - 2024*

- Built an end-to-end LLMOps telemetry collector tracking token consumption, inference latency distributions, cost attribution, and prompt failure rates across production model invocations.
- Designed consolidated operational Grafana dashboards with automated alerting to monitor model drift and ensure SLA reliability.

Certifications & Specialized Training

2024	**AI Infrastructure and Operations Fundamentals,** NVIDIA (Deep Learning Institute)	*Online*
2024	**Foundations of Project Management,** Google Career Certificates	*Online*
2023	**Generative AI for Everyone**, DeepLearning.AI	*Online*
2023	**Mathematics for Machine Learning and Data Science,** DeepLearning.AI	*Online*
2023	**AWS Certified Cloud Practitioner,** Amazon Web Services (AWS)	*Online*
2023	**Machine Learning,** Stanford University	*Online*
2019	**Cloud Infrastructure, Kubernetes & OpenStack**, The Linux Foundation	*Online*

Education

Keio University

Tokyo, Japan

Ph.D. in Media and Governance (Internet Architecture & Distributed Systems) — *Oct. 2014*

- Dissertation: Dynamic P2P network topologies, swarm dynamics, and energy-efficient Content Delivery Networks (CDNs).
- Supervisors: Prof. Jun Murai (Internet Hall of Fame / WIDE Project), Prof. Rodney Van Meter (Quantum Networking Pioneer).

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

M.T. (Master of Technology in Electrical Engineering) — *Oct. 2004*

- Designed and benchmarked one of Indonesia's earliest IPv6 campus network backbones and core routing deployments.

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

S.Si. (Bachelor of Science in Physics) — *Oct. 2000*

- Researched data assimilation and numerical simulation in 1D hydrodynamic fluid systems (differential wave equations).